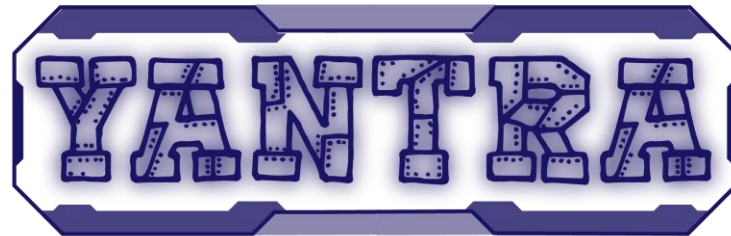


Cad & reverse engineering



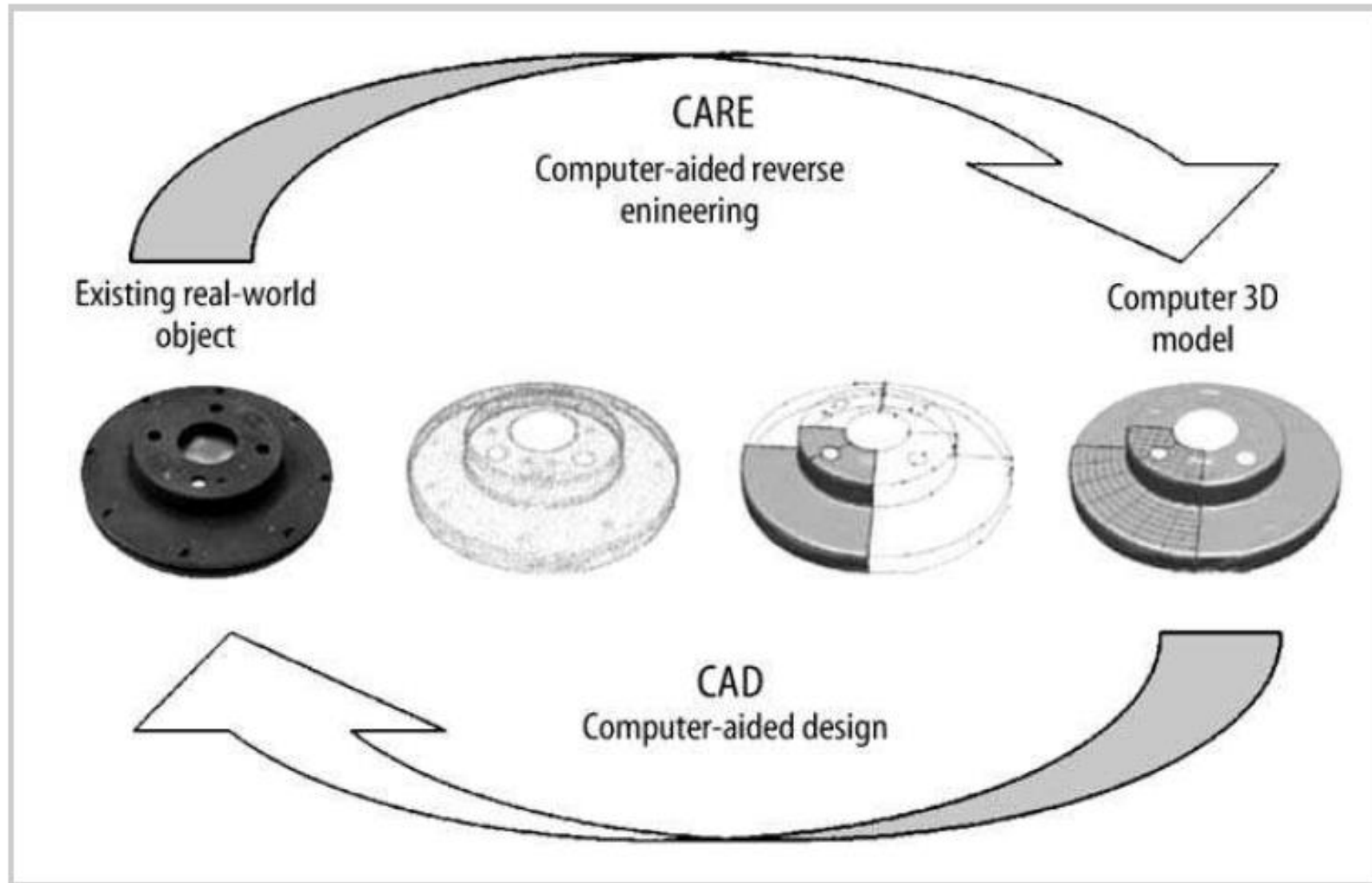
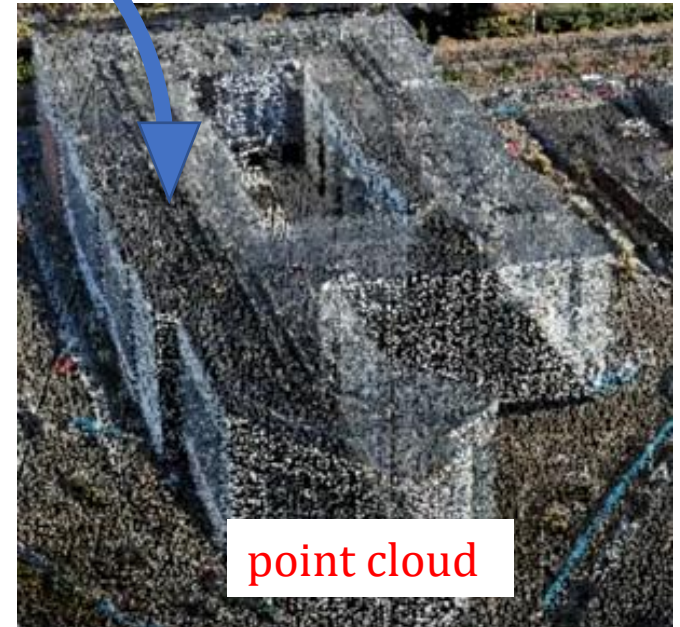
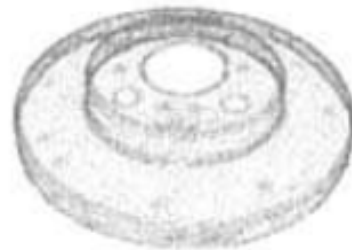


Figure 2.1. Computer-aided reverse engineering (CARE) process

1. The first step in the CARE process is to **make measurements at points** along the surface of the object.

Each point has an **x, y, and z coordinate** locating the point in 3-D space. For a given object, a CARE system will measure **hundreds** or even **thousands** of such **points** depending on the nature of the object and the type of CARE system.

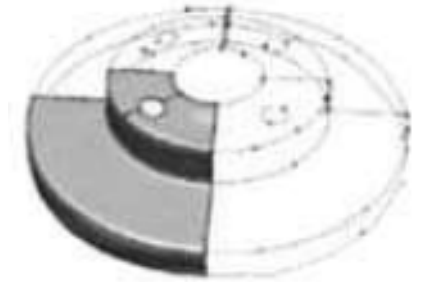
2. The collection of these points is known as a **point cloud**; an example appears in the second picture from the left in Figure.



point cloud

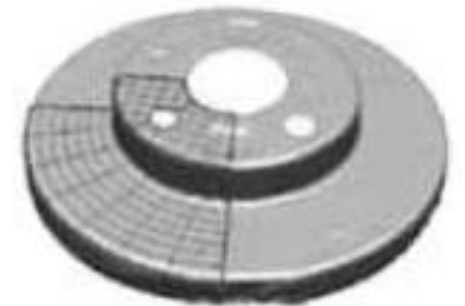
In most applications, the point cloud is a sufficient description of the object. However, higher levels are possible as the remaining two pictures on the right show.

3. The third picture from the left is a feature description of the object, in which the CARE system has detected **surface edges** and **creases** from the point cloud data.



4. The final picture on the right is a full and **complete CAD** description of the object.

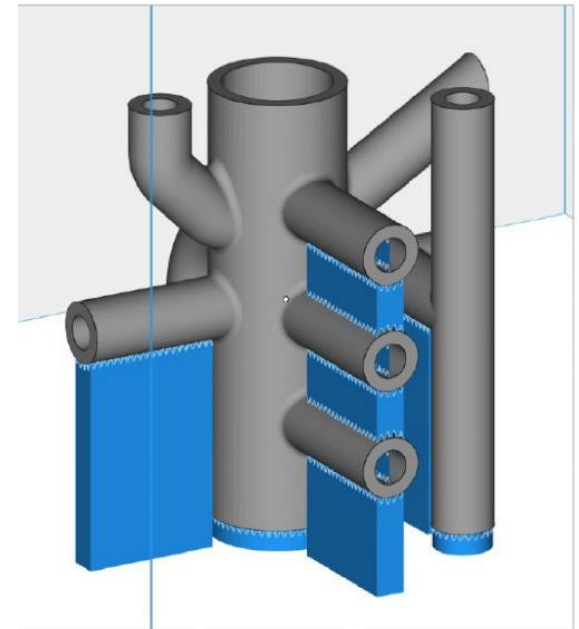
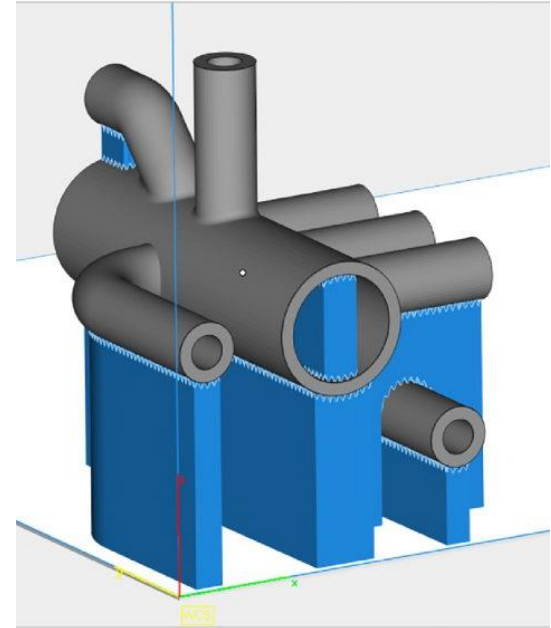
For this description, the CARE system uses the point cloud and the detected features to fit surfaces for modeling the entire geometry of the object.



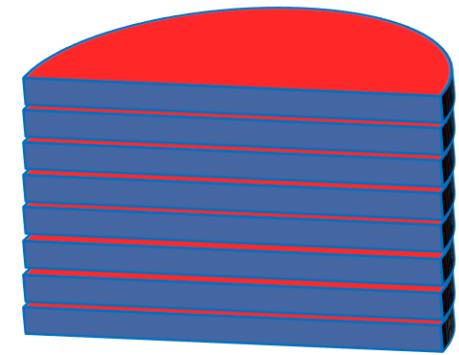
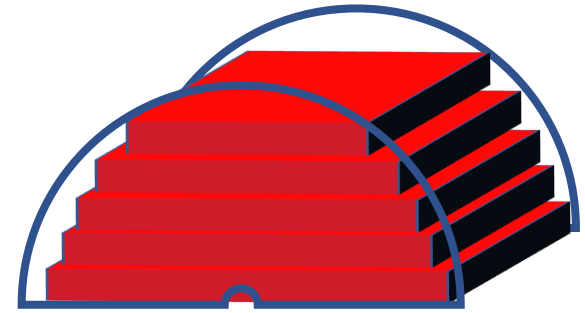
Data Processing for Additive Manufacturing Technology

Part Orientation

- **Choice of part orientation** is important as the orientation within the processing chamber influences the **build time**, **part resolution** and **surface finish**.
- **Higher resolutions** can be obtained in curved surfaces by orienting them in the **horizontal plane** (normal to the direction of layer deposition).

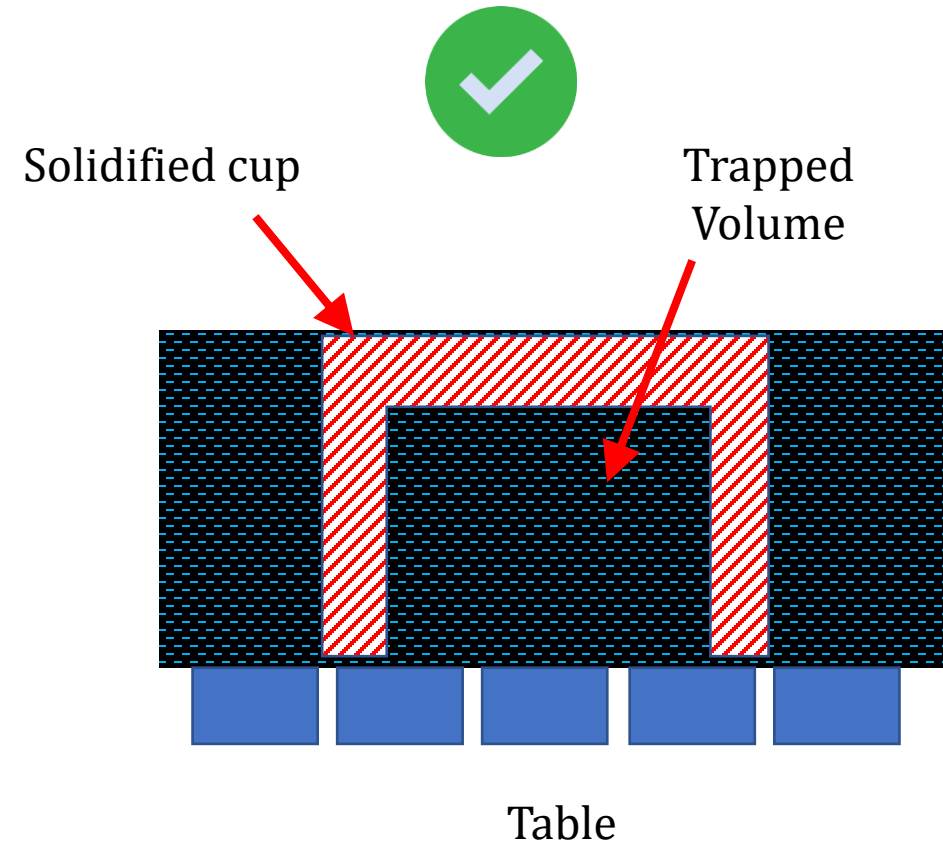
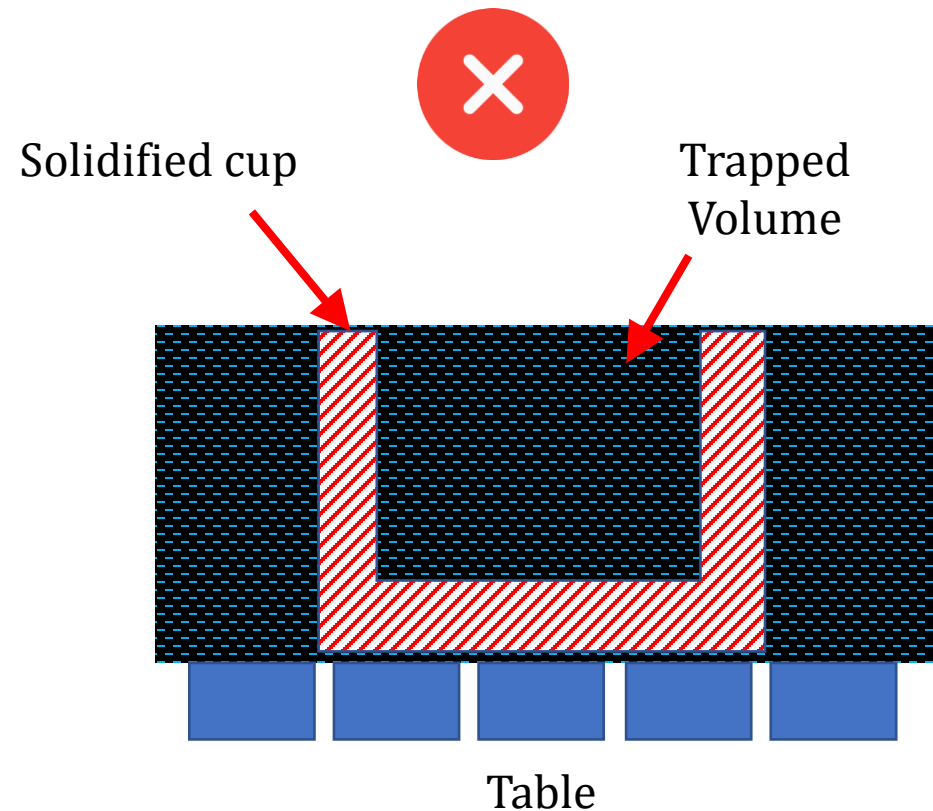


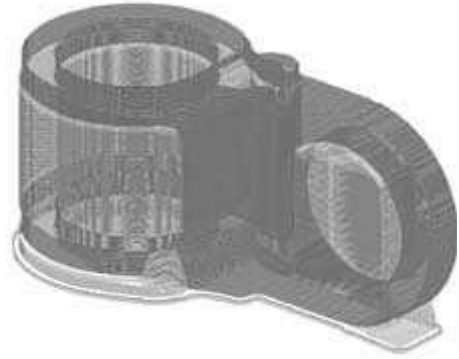
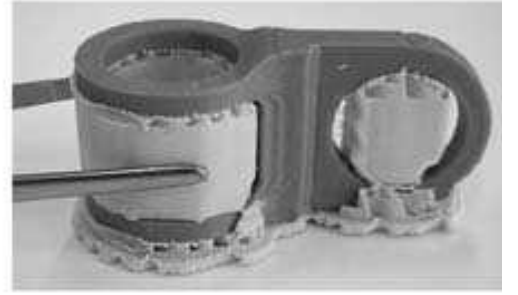
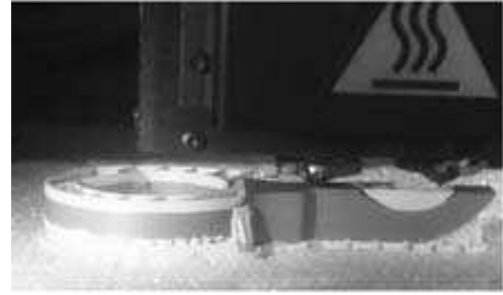
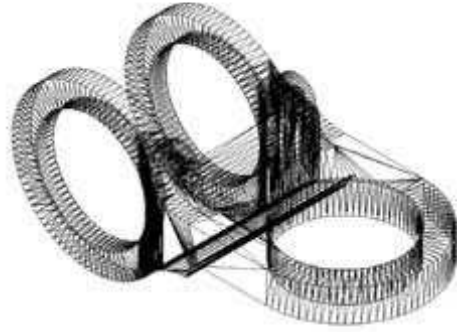
- Figures show clearly how the 'stair-step' appearance can be **avoided** by a proper choice of orientation.
- In cases where formation of **curved surfaces** in the direction of layer deposition is unavoidable, multiple layer thickness within a building cycle is feasible.
- **Surfaces with large slants** should be developed by using **thinner layers** to reduce the 'stair-step' effect.



- The orientation can also affect the **trapped volume** in case of liquid-based GMP's.
- The trapped volume is represented by that space which holds liquid that is completely separate from the liquid in the main vat. Figure shows how a correct choice of orientation can eliminate (or reduce) the problem.

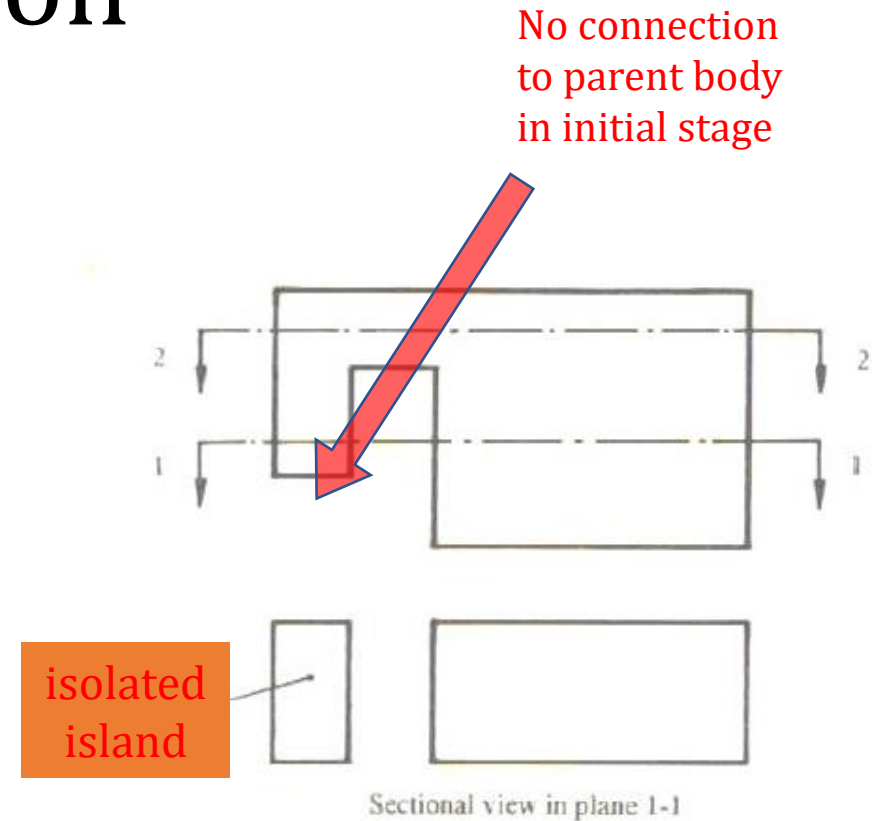
** This is in case of SLA*



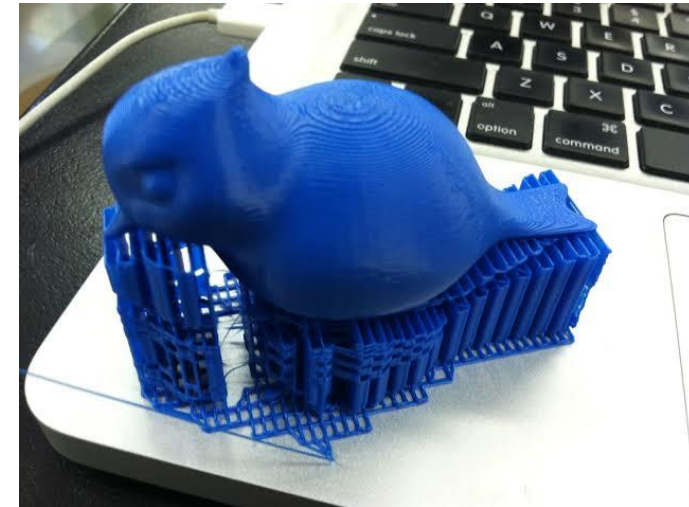
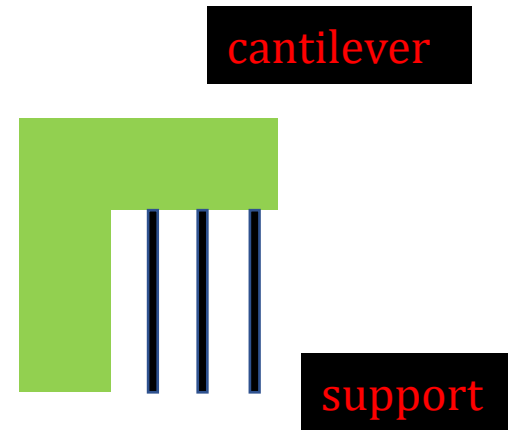


Support generation

- While slicing the CAD model into layers isolated islands may be produced as shown in Figure.
- The sectional view in plane 1-1 shows an **isolated island** which belongs to a projection from the main object.
- The connection of the projection to the parent body is from the top and while generating the shape by a GMP it will be **built later**.
- Thus it becomes essential to design a support for the isolated islands to prevent their **fall under gravity**, as they are created if the process is liquid-based like Stereolithography.
- When the whole object is formed the extra supports are removed.

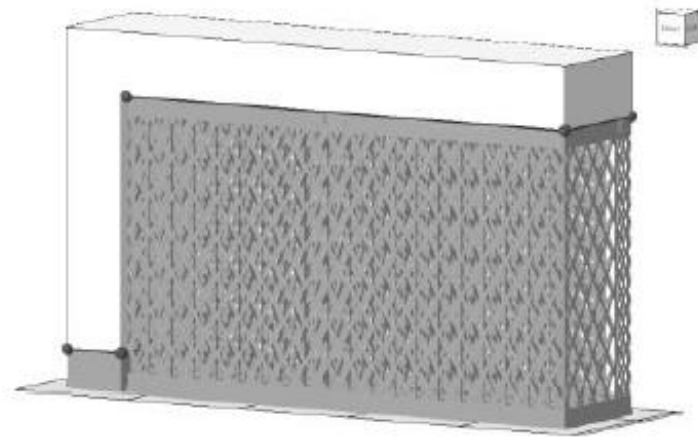


- Due to similar reasons, supports are essential for long **cantilevered** projections also.
- Though isolated islands are not formed, the thickness of the projection may be too **thin to support** the weight of the cantilever.
- Thus, supports in a GMP system are analogous to **job holding devices** for conventional machining.
- In addition to preventing the fall of isolated islands, supports are generally provided **to hold the main part body also**.
- **In future** suitable materials for GMP may be developed to eliminate the need of supporting the main object.
- At present supports are essential to hold the material during operation even if the component is devoid of cantilevers and projections.

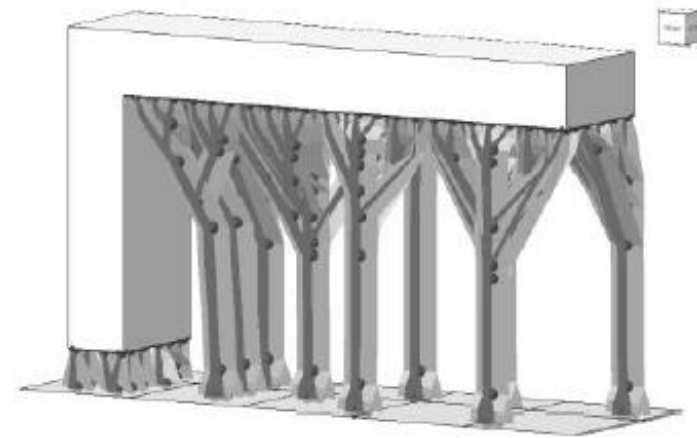


**Supports
holding
the job**

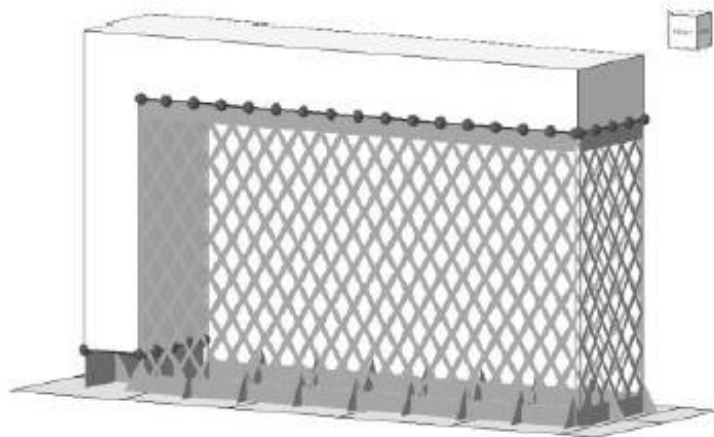
Figure shows different types of supports.



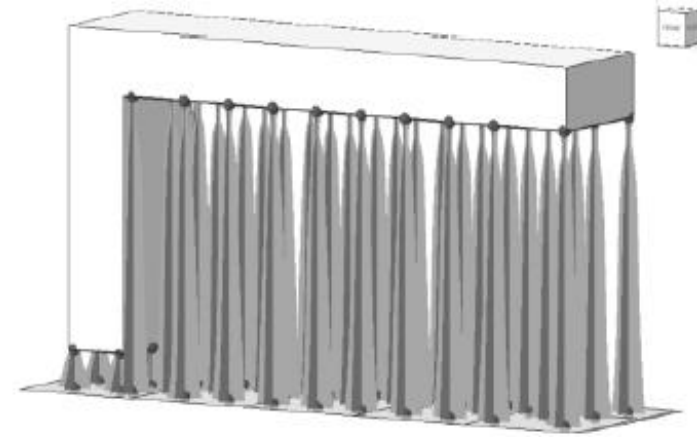
(a)



(b)



(c)

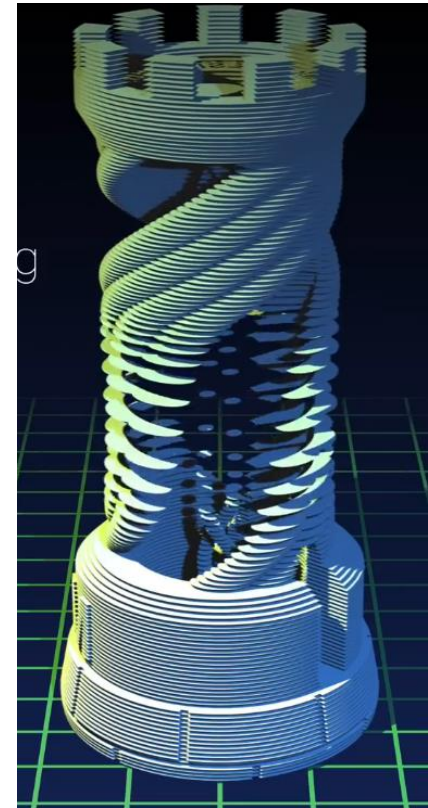


(d)

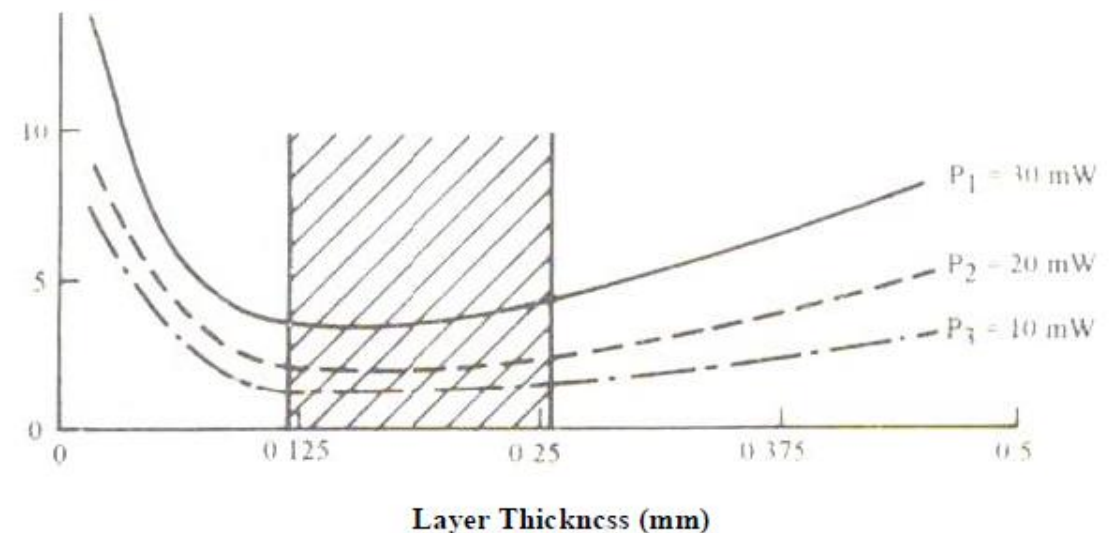
Figure 2. Types of support structure: Block-Support (a), Tree-like-Support (b), Polyline-Support (c) and Bar-Support (d)

Model Slicing

- The slice program converts the **three-dimensional object** in the STL file into **two-dimensional cross-sections**.
- The slice axis is defined as the **normal** to the plane created by slicing and this is also the build direction while creating the part by GMP.
- The **thickness** of slice dictates the **texture**, **accuracy** and **build time**.
- The layer thickness is normally in the range 0.0625 mm to 0.75 mm.



- It is, however, **not correct** to assume that **using thicker layers** (and reducing the number of layers) leads to **reduced build time** in all cases.
- In many processes the **speed of scanning** of the activating element (laser beam in many processes) **depends greatly on the layer thickness**.
- So, the **time** required for creating individual layers **increases** greatly when large thickness is used. Figure 1.10 shows the typical characteristics of how the build time changes when the layer thickness is gradually increased for three different power levels of the beam used. It is seen that the range **0.125 mm to 0.25 mm** is the optimum irrespective of the beam power.



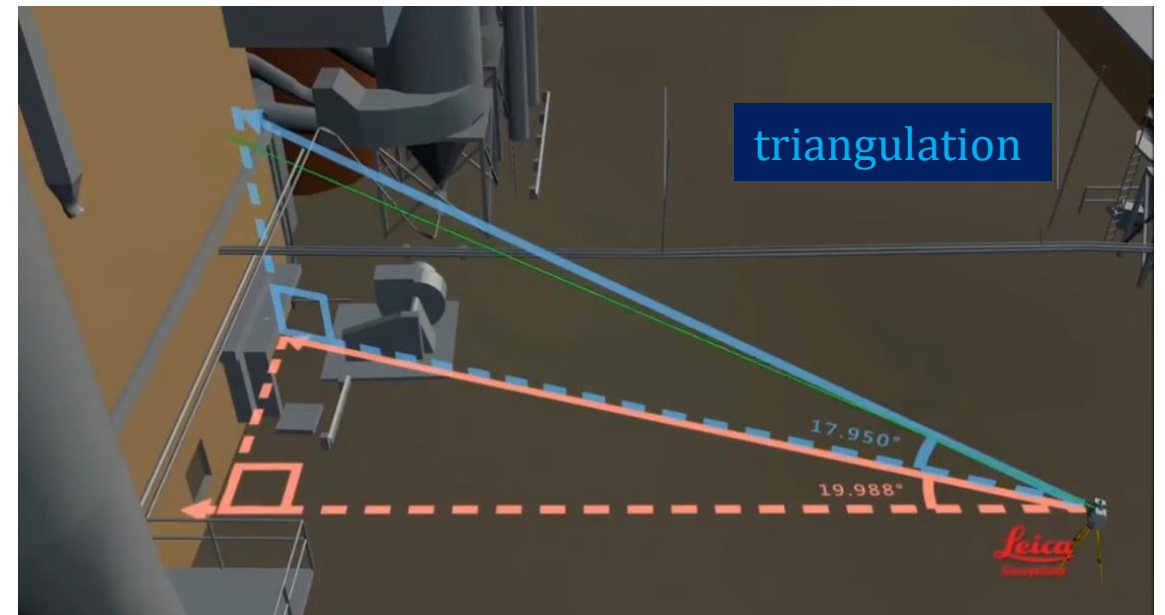
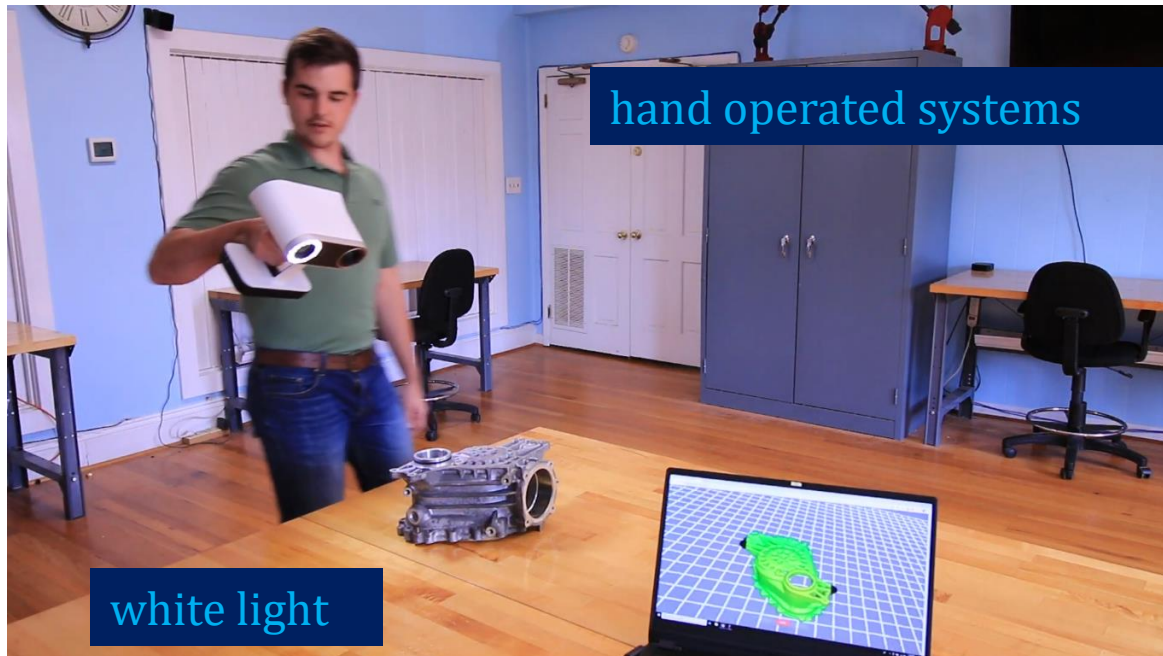
Surface digitization

- Technologies used commercially for the measurement of the surface of objects with micro to macro sizes can be divided fundamentally into **two groups**:
 - systems based on **laser scanning** and
 - systems based on **white light projection**.
-
- The used equipment is different; however they are based on the same principle: **triangulation**
 - **Laser scanning** systems employs lasers to project a spot, a line, multiple lines, or patterns onto a surface, whereas a light sensor, usually a camera, acquires the scene.
 - The three elements laser, light sensor and object surface form a triangle. When the geometrical disposition of the **laser** and the **light sensor** are known, the **distance** of the object surface to the laser scanning device can be easily determined by **triangulation**

- To measure surface areas the laser spot, line, multiple lines or pattern have to move over the area (i.e. scan the surface).
- For this process, different methods can be used, e.g. mirrors systems, electro-mechanical systems, hand operated systems.

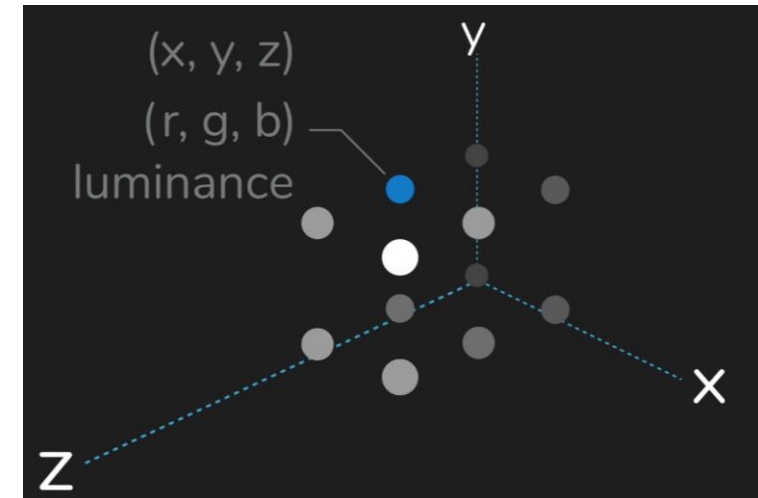
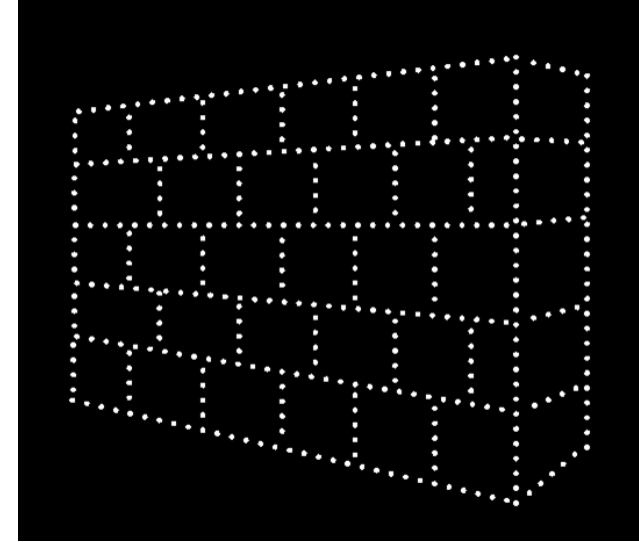


- 3D measurement systems based on **white light** employ projectors instead of laser light sources, to project light patterns onto the surface.
- The measurement principle remains the same: triangulation. A triangle is formed by projector, camera and object

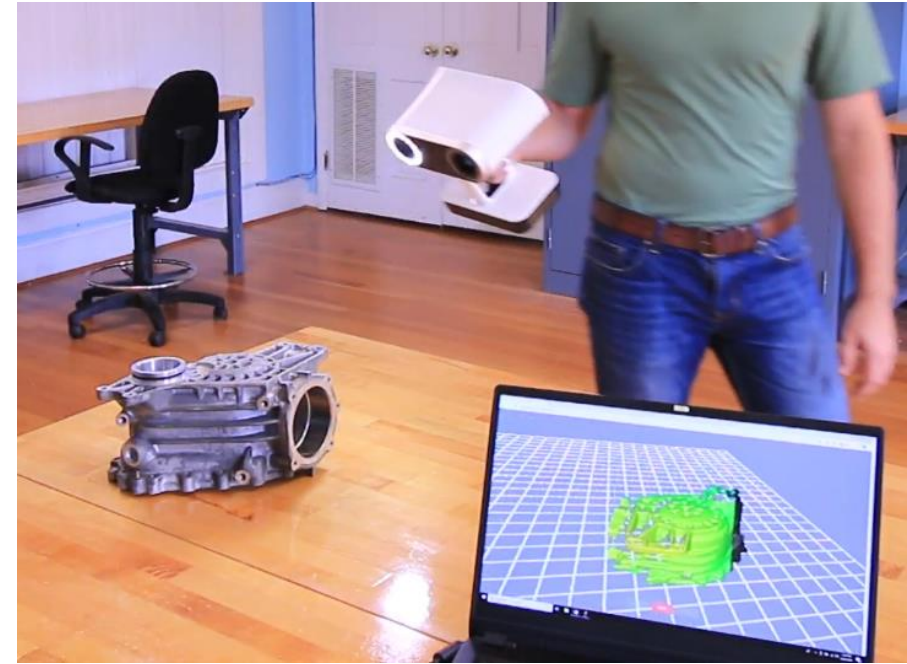


Surface generation from point cloud

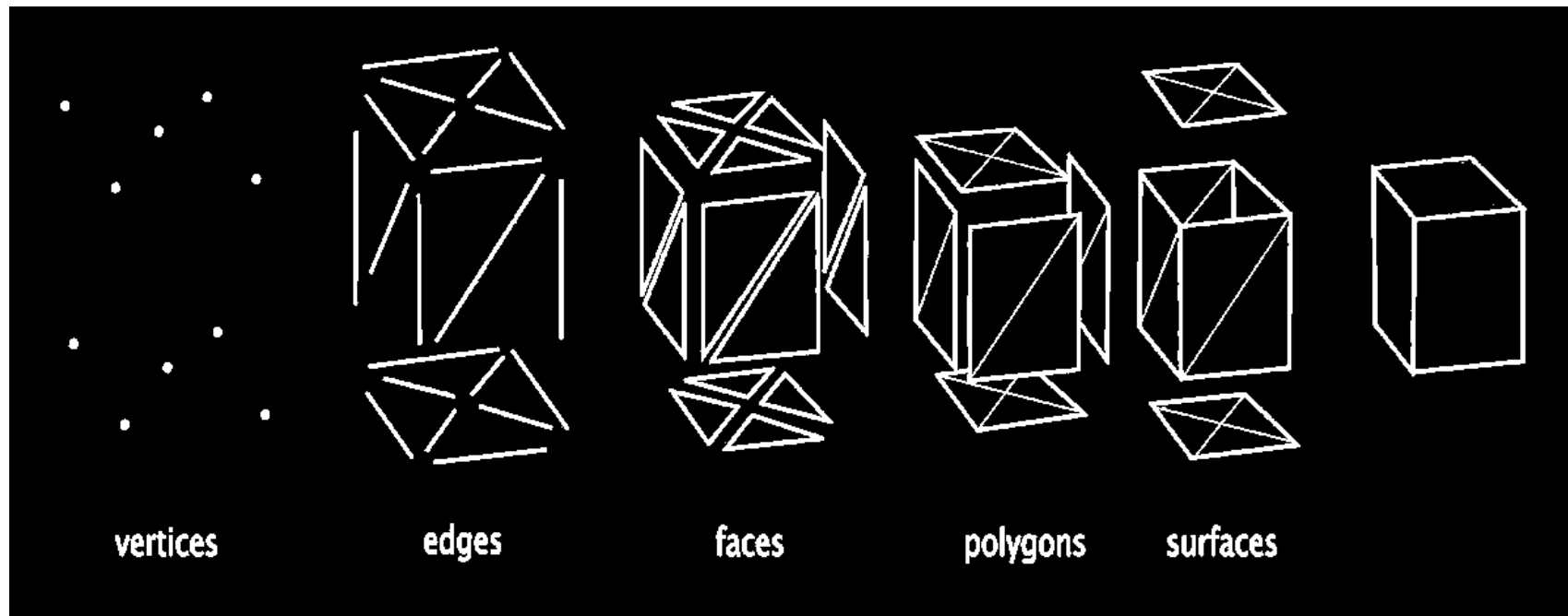
- A **point cloud** is a set of data points in some coordinate system. In a three-dimensional coordinate system, these points are usually defined by X, Y, and Z coordinates, and often are intended to represent the external surface of an object.

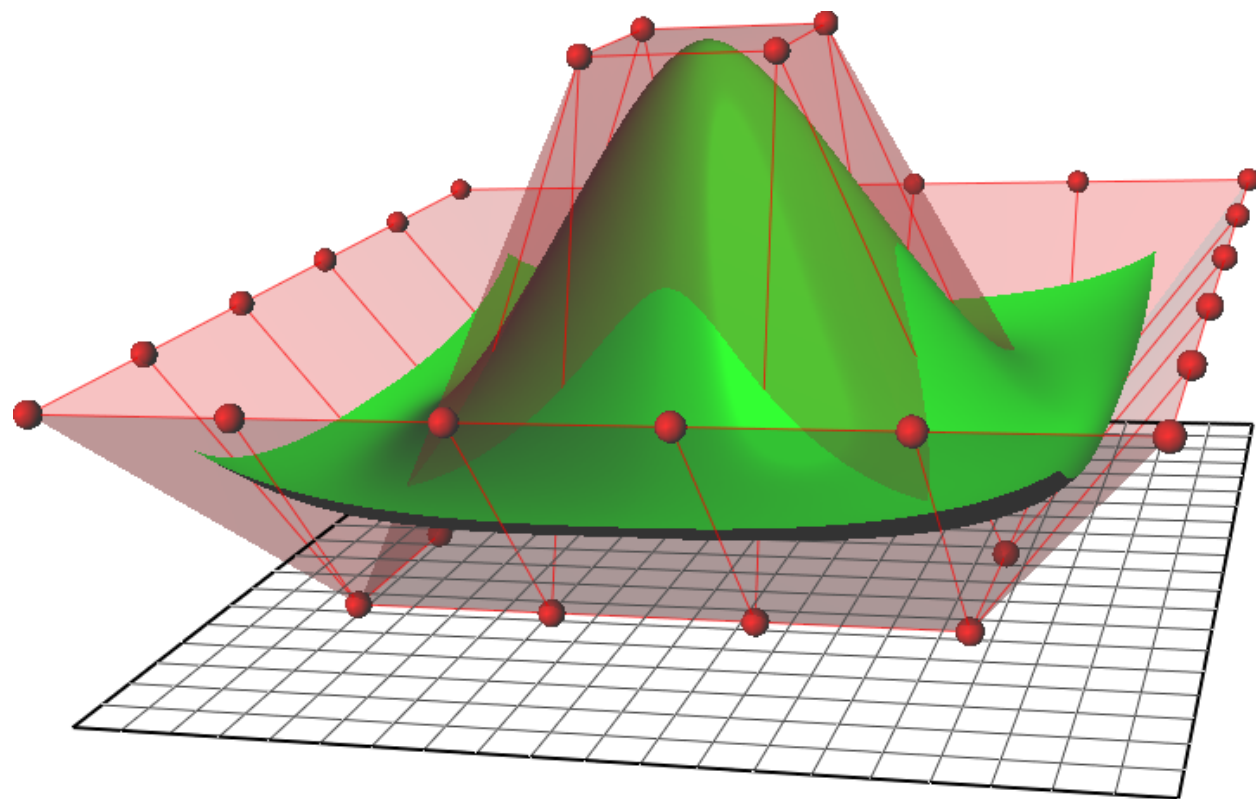


- Point clouds may be created by **3D scanners**.
- These devices measure a large number of points on an object's surface, and often output a point cloud as a data file.
- The point cloud represents the set of points that the device has measured.
- As the output of 3D scanning processes, point clouds are used for many purposes, **including to create 3D CAD models for manufactured parts, metrology/quality inspection, and a multitude of visualization, animation, rendering and mass customization applications.**



- While point clouds can be directly rendered and inspected, usually point clouds themselves are generally **not directly usable** in most 3D applications, and therefore **are usually converted to polygon** mesh or triangle mesh models, **NURBS** surface models, or CAD models through a process commonly referred to as **surface reconstruction**





Non-uniform rational B-spline

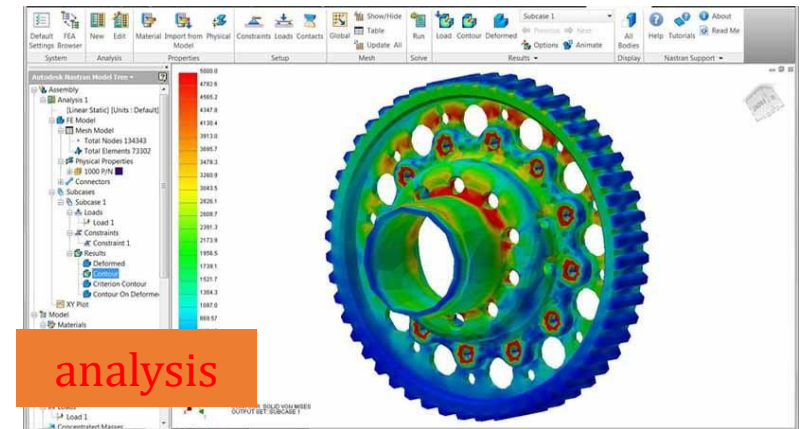
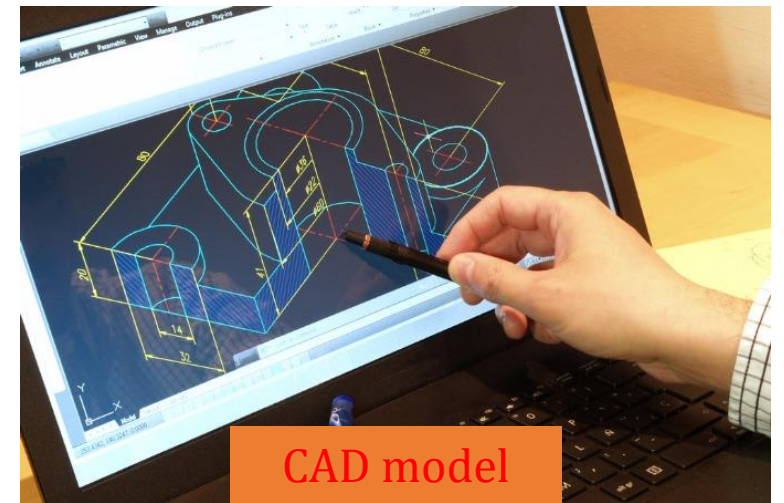
Surface modification-data transfer to solid models

- CAD data exchange is a modality of data exchange used to translate data between different Computer-aided design (CAD) authoring systems or between CAD and other downstream CAx systems.
- Many companies use different CAD systems internally and exchange CAD data with suppliers, customers and subcontractors.



Transfer of data is necessary so that, for example, **one** organization can be developing a CAD model, while **another** performs analysis work on the same model; at the same time a **third** organization is responsible for manufacturing the product.

The CAD systems currently available in the market **differ** not only in their application aims, user interfaces and performance levels, but also in **data structures** and **data formats**[3] therefore accuracy in the data exchange process is of paramount importance and robust exchange mechanisms are needed.



- The exchange process targets primarily the **geometric information** of the CAD data
- but it can also target other aspects such as metadata, knowledge, manufacturing information, tolerances and assembly structure.
- There are three options available for CAD data exchange: **direct model translation, neutral file exchange and third-party translators.**